UNIVERSITY OF PIRAEUS

DEPARTMENT OF DIGITAL SYSTEMS
MSc Program: Information Systems and Services
Specialization: Big Data and Analytics
Course: Big Data Programming and Infrastructures: Python and Cloud Computing

Prediction of Drug-Induced Autoimmunity (DIA)

Student Team:
Marina Nouchou — Student ID: 2423
Panagiotis Athanasopoulos — Student ID: 2402

Academic Year: 2024-2025

Contents

1. Introduction

Predicting Drug-Induced Autoimmunity (DIA) is a critical challenge in drug toxicology, because rare but serious autoimmune reactions can lead to drug withdrawal or severe adverse effects. The dataset [14] contains 597 drugs (148 positive, 449 negative) and constitutes a binary classification problem with substantial class imbalance (approximately 25% positive).

Recent studies have reported strong performance on this dataset. In particular, the InterDIA platform [10] proposed an integrated and interpretable DIA prediction framework based on molecular physicochemical properties. The model used techniques such as:

- Multi-strategy feature selection
- Advanced ensemble resampling to address class imbalance
- Easy Ensemble Classifier, achieving AUC=0.8836 (CV) and AUC=0.8930 (external test set)
- SHAP for mechanistic interpretation of predictions (properties such as lipophilicity, polarity and charges)
- Development of a web platform for rapid evaluation (Streamlit)

InterDIA demonstrated that appropriate molecular characterization combined with interpretable models can enable early immunotoxicity prediction, even when clinical diagnosis is complex. SHAP waterfall plots were also used to understand the role of each feature at molecule level.

Another comparative study by Delfiero et al. [4] compared XGBoost, CatBoost and Gradient Boosting. CatBoost achieved the highest ROC-AUC, while XGBoost provided the best balance between precision and recall. All three models achieved 87-89% accuracy, reinforcing the need to use metrics such as F1-score and MCC.

This problem requires specialized approaches because of class imbalance. Techniques such as SMOTE, SMOTEENN, EasyEnsemble, cost-sensitive learning, threshold tuning and stacking ensembles have been used to improve recall and precision. At the same time, feature selection and enrichment through RDKit, MOE or molecular fingerprints, the use of GNNs and transformer-based models, and calibration with Bayesian optimization have contributed to improved performance.

Objective: The aim of this work is to optimize DIA prediction models by examining and extending existing approaches. Through appropriate preprocessing, feature selection, imbalance-handling techniques and combined models (XGBoost, Random Forest, Logistic Regression), the study targets an F1-score above 75%, while emphasizing interpretability and prediction reliability through SHAP, confusion matrices and calibrated probabilities.

2. Data

The dataset used in this study comes from the UCI Machine Learning Repository and concerns prediction of autoimmunity caused by pharmaceutical compounds (Drug-Induced Autoimmunity - DIA) [14]. It contains 597 pharmaceutical molecules described by 195 continuous molecular descriptors extracted with the RDKit library.

Data Structure

Each sample (drug molecule) is represented by one row containing:

- 195 descriptors: Numerical features describing molecular, physicochemical, topological, electronic and structural properties of the molecule (e.g. MolWt, TPSA, BalabanJ, etc.).
- SMILES: String representation of the chemical structure (not used directly as model input).
- Label: Binary target label - 1 for DIA-positive molecules and 0 for DIA-negative molecules.

Training/Test Split

The dataset is already divided into:

- Training Set: 477 samples (118 positive, 359 negative).
- Test Set: 120 samples (30 positive, 90 negative).

The positive-to-negative ratio is approximately 1:3 in each set, indicating a class-imbalance problem that must be considered during modeling.

Data Completeness

All fields were checked for missing values (null / NaN), and no missing values were found. All fields contain complete and valid records.

Class Distribution

- Label = 1 (DIA-positive): 118 samples
- Label = 0 (DIA-negative): 359 samples

This corresponds to 24.7% positive and 75.3% negative samples, which is one of the main reasons for prioritizing performance metrics such as F1-score and Recall instead of Accuracy alone.

Analysis Objective

The objective is to train classifiers that predict Label (0 or 1) from the 195 descriptors in order to identify drugs that may induce autoimmunity. Emphasis is placed not only on predictive accuracy, but also on interpretability through feature analysis.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the main characteristics of the DIA dataset, identify potential asymmetries, determine the need for normalization and prepare the dataset appropriately for modeling.

3.1. Dimensions and Basic Statistics

The original dataset contains 597 drug molecules, each described by 195 molecular descriptors (RDKit descriptors). It also includes SMILES and Label fields.

- Training set: 477 records (118 positive, 359 negative)
- Test set: 120 records (30 positive, 90 negative)

No missing or invalid data were detected. All descriptors are numerical and therefore suitable for machine-learning algorithms.

3.2. Class Distribution

- DIA-negative (Label=0): 359 samples (75%)
- DIA-positive (Label=1): 118 samples (25%)

This imbalance justifies using metrics such as F1-score, Recall and ROC AUC instead of Accuracy alone, which can be misleading.

3.3. Descriptive Statistics and Normalization

Descriptive statistics obtained through pandas.describe() reveal substantial heterogeneity in descriptor scales. The following plots show the distributions of key statistics:

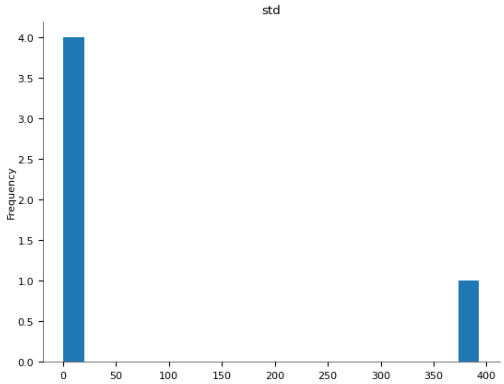
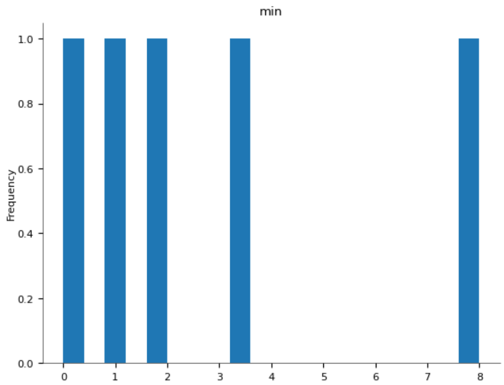


Figure 1: Distribution of standard deviation (std) and minimum value (min) across descriptors.



The variance is large, indicating asymmetry.

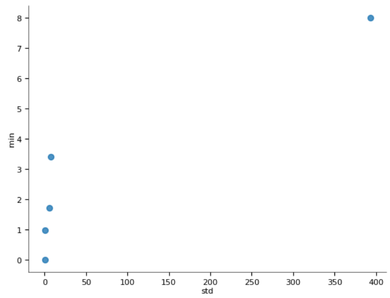


Figure 2: Scatter plots.

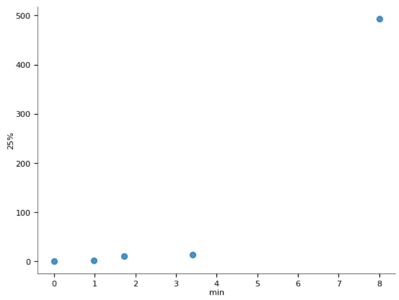


Figure 3: Mean and count of selected descriptors. BertzCT exhibits exceptionally high values.

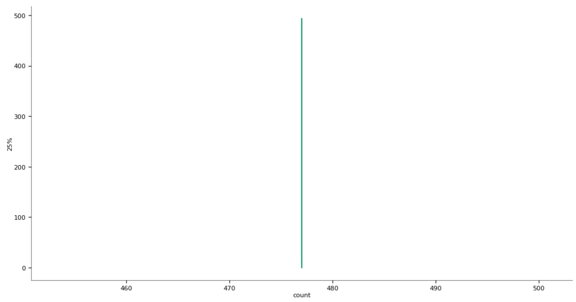
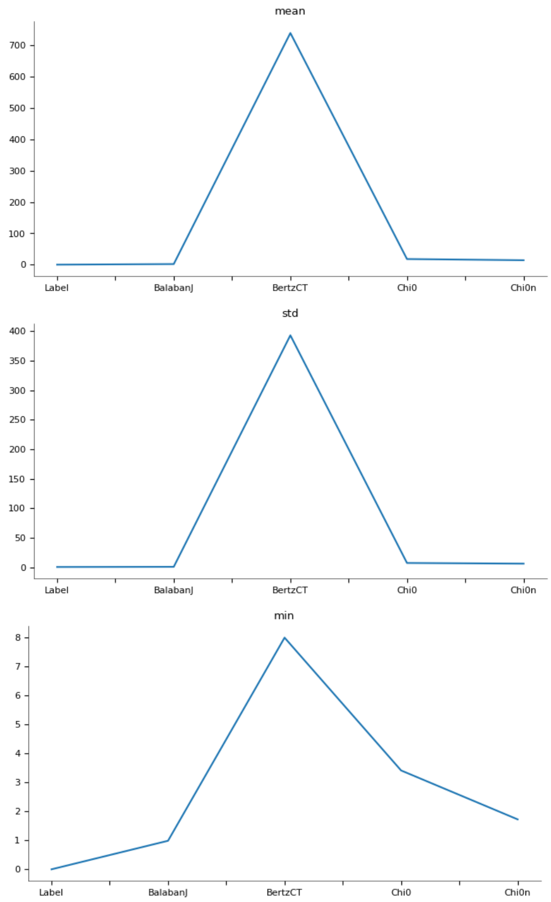


Figure 4: Standard deviation and minimum values. Chi0 and Chi0n are very close to 0, indicating limited information.



Scatter plots of min vs std and min vs 25% indicate the presence of outliers in specific features.

3.4. Comparative Overview of Descriptors

To identify descriptors with extreme values or redundant information, line plots of basic statistics were examined for selected descriptors (BalabanJ, BertzCT, Chi0, Chi0n, etc.).

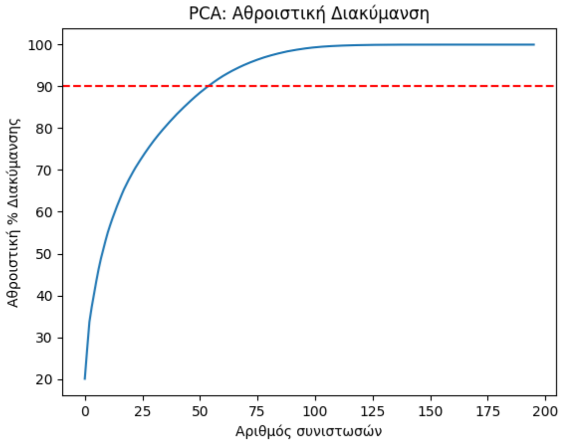


Figure 5: The 25th percentile confirms the concentration range of values. On the right, cumulative variance shows the effectiveness of PCA.

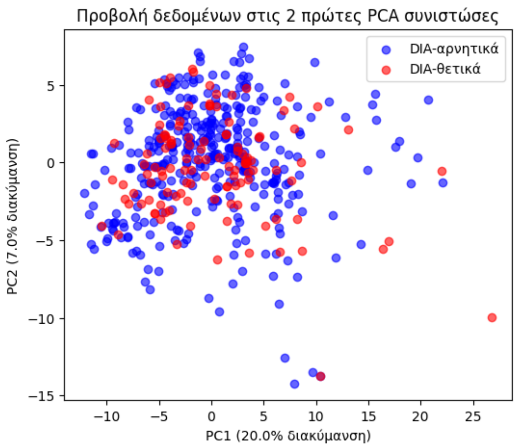
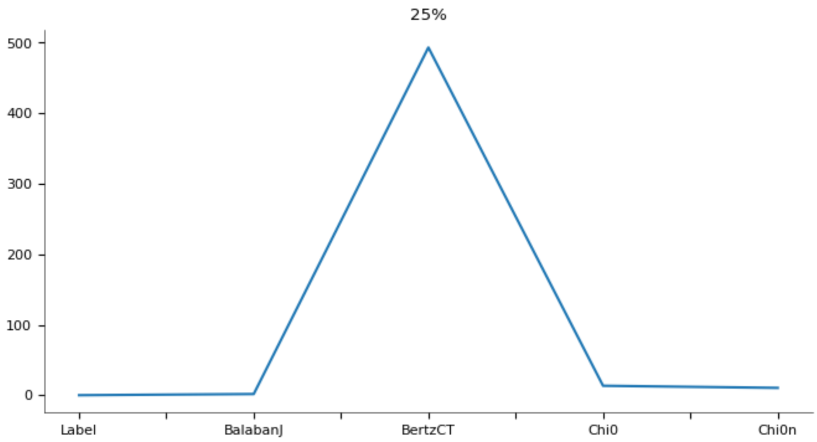


Figure 6: Projection of samples onto the first two PCA components. No clear separation between classes is observed.



3.5. PCA (Principal Component Analysis)

For dimensionality reduction and visual inspection, PCA was applied to the descriptors. The cumulative variance curve shows:

- 90% of the information is retained with the first ~75 components.
- 95% is covered with fewer than 100 components.

Although PCA successfully compresses the information, the visualization does not indicate linear class separation. Therefore, nonlinear classification algorithms (such as XGBoost, SVM, CatBoost, etc.) are required.

4. Methodology

The DIA prediction system was developed through sequential stages covering data preprocessing, incorporation of chemical information through Tanimoto graph representation, class-imbalance handling, feature selection, model training and final evaluation.

4.1. Preprocessing and Normalization

Initially, MinMax normalization was applied to all RDKit descriptors so that their values were mapped to [0,1]. Zero-variance features, duplicate columns and highly correlated features (corr > 0.95) were removed to reduce dimensionality and redundancy.

4.2. Feature Enrichment: Tanimoto Graph Embeddings

To incorporate molecular chemical structure, Tanimoto-based similarity graphs were constructed from ECFP6 fingerprints [13]. Laplacian Eigenmaps embeddings (dim=64-96) [1] were extracted from these graphs and graph smoothing was applied to the descriptors [16]. In addition, the top ECFP bits were selected using the chi-square statistic [7] and appended to the final feature representation.

4.3. Handling Class Imbalance

- SMOTE for oversampling the minority class (ratio=0.8) [3]
- Scaffold-aware Cross-Validation based on Murcko scaffolds to avoid data leakage [12]
- No class weights when SMOTE is used, avoiding double weighting [2]
- Dual-threshold classification for uncertain predictions: two thresholds with minimum coverage and Fβ performance [6]
- Isotonic calibration of probabilities to reduce overconfidence [15]
- Seed averaging: 5 repetitions per model with different random seeds and mean predicted probabilities [5]

4.4. Feature Selection (Feature Engineering)

- SHAP values based on XGBoost for feature-contribution assessment [11]
- Recursive Feature Elimination (RFE) for gradual feature removal [9]
- Genetic Selection using DEAP (evolutionary optimization for F1) [8]

The final feature set consisted of the union of features selected by the above methods.

4.5. Models and Training

Three main models were trained:

- LogisticRegression (elasticnet penalty): used with grid search and isotonic calibration.
- RandomForestClassifier: 1,000+ trees, optimized hyperparameters and isotonic-regression calibration.
- XGBClassifier: GPU-accelerated, tuned with RandomizedSearchCV and calibrated.

All models were evaluated with 5-fold Scaffold Cross-Validation using F1-score, Recall, Precision and ROC-AUC. Final predictions were based on calibrated probabilities and a dual-threshold uncertainty-abstention strategy.

5. Results

This section presents the results of the three main models: XGBoost, Random Forest and Logistic Regression. Evaluation includes both seed-averaged calibrated models and single-fit models, together with ROC/PR curves and confusion matrices.

5.1. Comparative Metrics Table

Model	Version	Accuracy	Precision	Recall	F1-score	ROC AUC
XGBoost	Seed-Avg	0.8917	0.8696	0.6667	0.7547	0.9274
	Single-Fit	0.8750	0.8261	0.6333	0.7170	0.9044
Random Forest	Seed-Avg	0.8917	0.8148	0.7333	0.7719	0.9326
	Single-Fit	0.8750	0.8000	0.6667	0.7273	0.9357
Logistic Regression	Seed-Avg	0.8583	0.8824	0.5000	0.6383	0.8770

Model	Version	Accuracy	Precision	Recall	F1-score	ROC AUC
	Single-Fit	0.8583	0.8824	0.5000	0.6383	0.8770

Table 1: Comparative metrics for XGBoost, Random Forest and Logistic Regression (Test Set)

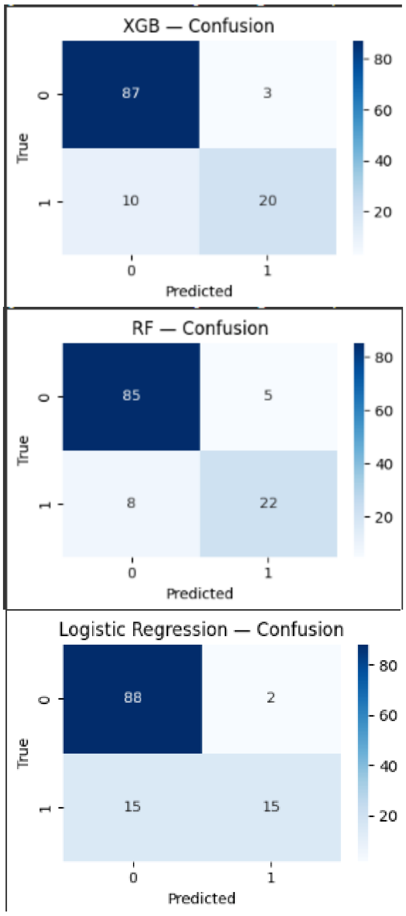
5.2. Results Analysis

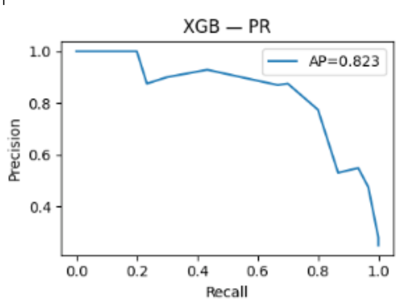
XGBoost achieved high Precision (0.87) and ROC AUC (0.927), indicating strong discrimination between classes. However, its Recall (0.6667) was lower than that of Random Forest, indicating the presence of false-negative predictions.

Random Forest achieved the best balance between Precision and Recall, with Recall = 0.7333 and F1-score = 0.7719. ROC AUC = 0.933 was the highest among the three seed-averaged models, confirming its strong overall performance.

Logistic Regression, although simpler, achieved very high Precision (0.8824) but low Recall (0.5000), corresponding to more false negatives. Its overall discrimination remained satisfactory (ROC AUC = 0.877), making it a reasonable baseline model.

5.3. Confusion Matrices





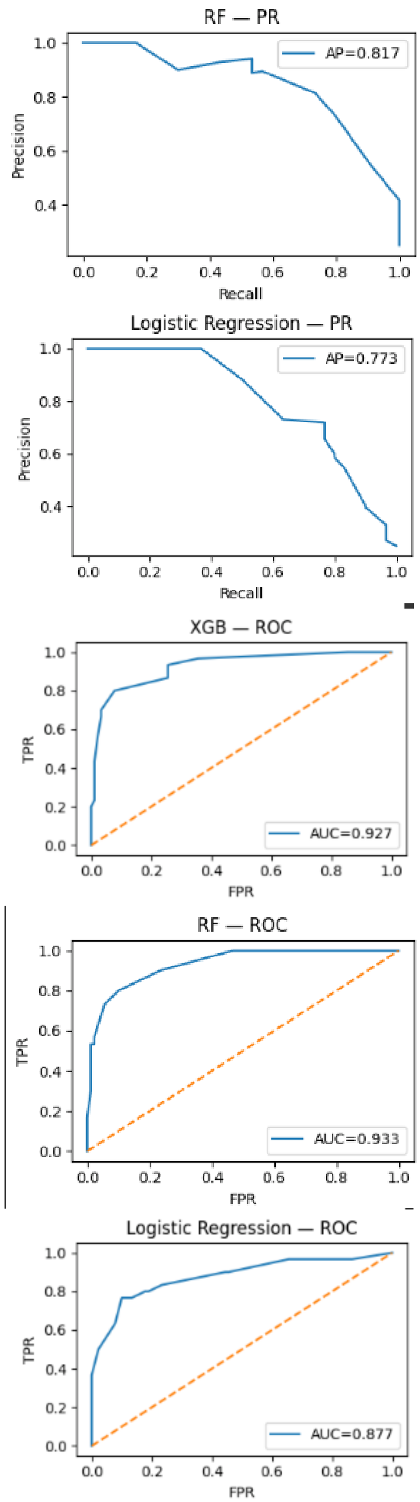


Figure 7: Confusion Matrices for XGBoost, Random Forest and Logistic Regression, respectively.

- XGBoost had 10 False Negatives (FN), reducing recall.
- Random Forest had only 8 FN and balanced True Positives (TP).
- Logistic Regression had 15 FN and 15 TP, indicating low sensitivity.

5.4. Precision-Recall and ROC Curves

Figure 8: Precision-Recall curves for XGBoost (AP=0.823), RF (AP=0.817) and LR (AP=0.773).

Figure 9: ROC curves for XGBoost (AUC=0.927), RF (AUC=0.933) and LR (AUC=0.877).

The XGBoost PR curve shows high precision across much of the recall range, while Random Forest maintains a better balance. Logistic Regression shows a sharper decline in precision as recall increases, indicating lower reliability at high coverage.

The Random Forest ROC curve is more 'square-shaped' and closer to the ideal point (0,1), confirming that it is the strongest model on this dataset.

6. Conclusions

This study aimed to develop and evaluate classification models for detecting positive DIA cases in an imbalanced molecular dataset. Particular emphasis was placed on class-imbalance handling, prediction reliability and comparative analysis of different algorithms.

Model Performance: The evaluated models were XGBoost, Random Forest and Logistic Regression, using techniques such as SMOTE, seed averaging, isotonic calibration and dual-threshold classification.

- Random Forest achieved the best balance between Precision and Recall, with F1-score = 0.7719 and ROC AUC = 0.933. Its performance indicates stronger ability to identify positive cases without a substantial increase in false positives.
- XGBoost, although it achieved high Precision = 0.8696, had slightly lower Recall = 0.6667, supporting the view that it is more conservative when predicting positive cases.
- Logistic Regression, despite its simpler and more interpretable architecture, had substantially lower Recall = 0.5000, resulting in more missed positive cases. However, it achieved high Precision = 0.8824 and AUC = 0.877, making it a stable baseline.

General Observations: Seed Averaging helped stabilize results by reducing the effect of training stochasticity. Isotonic Calibration improved the reliability of output probabilities. The dual-threshold approach also enabled low-confidence regions to be defined, increasing the usefulness of the system in real-world scenarios.

Practical Significance: For applications in which identifying positive cases is critical (e.g. drug discovery or toxicological assessment), Random Forest is recommended as the most suitable model. Conversely, in settings where false-positive predictions should be minimized, XGBoost may be preferable.

Overall, the methodology demonstrates that even in imbalanced problems, careful preprocessing, strategic threshold handling and appropriate evaluation can produce models with strong generalization and high performance.

References

[1] M. Belkin and P. Niyogi, Laplacian Eigenmaps for Dimensionality Reduction and Data Representation, Neural Computation, vol. 15, no. 6, pp. 1373-1396, 2003.

[2] M. Buda, A. Maki, and M. A. Mazurowski, A Systematic Study of the Class Imbalance Problem in Convolutional Neural Networks, Neural Networks, vol. 106, pp. 249-259, 2018.

[3] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research, vol. 16, pp. 321-357, 2002.

[4] Delfiero et al., Comparative study of ML algorithms for predicting drug-induced autoimmunity using molecular descriptors, Jurnal Mandiri IT, vol. 14, no. 1, pp. 180-190, 2025.

[5] T. G. Dietterich, Ensemble Methods in Machine Learning, International Workshop on Multiple Classifier Systems, Springer, pp. 1-15, 2000.

[6] C. Elkan, The Foundations of Cost-Sensitive Learning, Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI), 2001.

[7] G. Forman, An Extensive Empirical Study of Feature Selection Metrics for Text Classification, Journal of Machine Learning Research, vol. 3, pp. 1289-1305, 2003.

[8] F.-A. Fortin, F.-M. De Rainville, M.-A. Gardner, M. Parizeau, and C. Gagné, DEAP: Evolutionary Algorithms Made Easy, Journal of Machine Learning Research, vol. 13, pp. 2171-2175, 2012.

[9] I. Guyon, J. Weston, S. Barnhill, and V. Vapnik, Gene Selection for Cancer Classification using Support Vector Machines, Machine Learning, vol. 46, no. 1-3, pp. 389-422, 2002.

[10] Huang et al., InterDIA: Interpretable Prediction of Drug-induced Autoimmunity through Ensemble Machine Learning Approaches, Toxicology, vol. 511, 2025.

[11] S. M. Lundberg and S.-I. Lee, A Unified Approach to Interpreting Model Predictions, Advances in Neural Information Processing Systems (NeurIPS), vol. 30, pp. 4765-4774, 2017.

[12] P. G. Polishchuk, T. I. Madzhidov, and A. Varnek, Estimation of the Size of Drug-like Chemical Space Based on GDB-17 Data, Journal of Computer-Aided Molecular Design, vol. 27, no. 8, pp. 675-679, 2013.

[13] D. Rogers and M. Hahn, Extended-Connectivity Fingerprints, Journal of Chemical Information and Modeling, vol. 50, no. 5, pp. 742-754, 2010.

[14] UCI Machine Learning Repository. Drug Induced Autoimmunity Prediction Dataset. archive.ics.uci.edu/dataset/1104/drug_induced_autoimmunity_prediction

[15] B. Zadrozny and C. Elkan, Transforming Classifier Scores into Accurate Multiclass Probability Estimates, Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 694-699, 2002.

[16] D. Zhou, O. Bousquet, T. N. Lal, J. Weston, and B. Schölkopf, Learning with Local and Global Consistency, Advances in Neural Information Processing Systems (NeurIPS), vol. 16, pp. 321-328, 2004.